tf.expand_dims

View source on GitHub (https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L391-L458)

Returns a tensor with a length 1 axis inserted at index `axis`.

```
tf.expand_dims(
    input, axis, name=None
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Extension types (https://www.tensorflow.org/guide/extension_type) • Import a JAX model using JAX2TF (https://www.tensorflow.org/guide/jax2tf) • Migrate `tf.feature_column`s to Keras preprocessing layers (https://www.tensorflow.org/guide/migrate/migrating_feature_columns) • Understanding masking & padding (https://www.tensorflow.org/guide/keras/understanding_masking_and_padding) • Working with RNNs (https://www.tensorflow.org/guide/keras/working_with_rnns)	• Integrated gradients (https://www.tensorflow.org/tutorials/interpretability/integrated_gradients) • Playing CartPole with the Actor-Critic method (https://www.tensorflow.org/tutorials/reinforcement_learning/actor_critic) • Generate music with an RNN (https://www.tensorflow.org/tutorials/audio/music_generation) • DeepDream (https://www.tensorflow.org/tutorials/generative/deepdream) • pix2pix: Image-to-image translation with a conditional GAN (https://www.tensorflow.org/tutorials/generative/pix2pix)

Given a tensor `input`, this operation inserts a dimension of length 1 at the dimension index `axis` of `input`'s shape. The dimension index follows Python indexing rules: It's zero-based, a negative index it is counted backward from the end.

This operation is useful to:

- Add an outer "batch" dimension to a single element.
- Align axes for broadcasting.
- To add an inner vector length axis to a tensor of scalars.

For example:

If you have a single image of shape `[height, width, channels]`:

```
>>> image = tf.zeros([10,10,3])
```

You can add an outer `batch` axis by passing `axis=0`:

```
>>> tf.expand_dims(image, axis=0).shape.as_list()
[1, 10, 10, 3]
```

The new axis location matches Python `list.insert(axis, 1)`:

```
>>> tf.expand_dims(image, axis=1).shape.as_list()
[10, 1, 10, 3]
```

Following standard Python indexing rules, a negative `axis` counts from the end so `axis=-1` adds an inner most dimension:

```
>>> tf.expand_dims(image, -1).shape.as_list()
```

```
[10, 10, 3, 1]
```

This operation requires that `axis` is a valid index for `input.shape`, following Python indexing rules:

```
-1-tf.rank(input) <= axis <= tf.rank(input)
```

This operation is related to:

- `tf.squeeze` (https://www.tensorflow.org/api_docs/python/tf/squeeze), which removes dimensions of size 1.
- `tf.reshape` (https://www.tensorflow.org/api_docs/python/tf/reshape), which provides more flexible reshaping capability.
- `tf.sparse.expand_dims` (https://www.tensorflow.org/api_docs/python/tf/sparse/expand_dims), which provides this functionality for `tf.SparseTensor` (https://www.tensorflow.org/api_docs/python/tf/sparse/SparseTensor)

Args	
`input`	A `Tensor`.
`axis`	Integer specifying the dimension index at which to expand the shape of `input`. Given an input of D dimensions, `axis` must be in range `[-(D+1), D]` (inclusive).
`name`	Optional string. The name of the output `Tensor`.

Returns
A tensor with the same data as `input`, with an additional dimension inserted at the index specified by `axis`.

Raises	
`TypeError`	If `axis` is not specified.
`InvalidArgumentError`	If `axis` is out of range `[-(D+1), D]`.

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